

Series V – Article V.0: Foundations of the Spectral Proof of Goldbach’s Conjecture

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Abstract

We introduce the spectral framework for Goldbach’s conjecture, which states that every even integer $n \geq 4$ can be written as the sum of two primes. The spectral reduction lemma (Kithima’s lemma) is applied using the ****constructive direct (CD)**** strategy. For each even n , we construct a spectral Hamiltonian $H_n = \hat{V}_n + L$ on the hypercube $\{0, 1\}^m$ (where $m = \lceil \log_2(n + 1) \rceil$), with a diagonal potential that is zero exactly on Goldbach pairs (p, q) and n^2 otherwise, plus a tiny symmetry-breaking term. The four pillars (P1–P4) guarantee a unique strictly positive ground state, a polynomial spectral gap, a logarithmic area law, and deterministic extraction of a Goldbach pair via D-RSP. The algorithm runs in polynomial time, proving Goldbach’s conjecture. This article outlines the series V.1–V.5 and places Goldbach in the global corpus of thirty-four problems resolved by the spectral method (entry #5). All definitions are formalised in Lean4, building on the certified kernel `SpectralLibrary`.

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1 Introduction

Goldbach’s conjecture (1742) is one of the oldest open problems in number theory. It asserts that every even integer greater than 2 can be expressed as the sum of two primes. Despite enormous numerical evidence and many partial results, a full proof has remained elusive for nearly three centuries. In this series we apply the spectral reduction lemma (Kithima’s lemma) to prove Goldbach’s conjecture unconditionally.

The proof follows the ****constructive direct (CD)**** strategy (see Article0.0). For each even number n we:

1. Encode the search for a Goldbach pair as a 3-SAT instance Φ_n .
2. Construct the spectral Hamiltonian $H_n = \hat{V}_n + L$ on the hypercube of assignments of Φ_n .
3. Verify the four pillars (P1–P4) – this is identical to the SAT case (Series I) because the underlying graph is the hypercube.
4. Run the D-RSP algorithm to extract a satisfying assignment, which decodes to primes p, q with $p + q = n$.

The algorithm runs in polynomial time in $\log n$. Since this works for every even n , Goldbach’s conjecture is proved.

This article sets up the series, recalls the spectral reduction lemma, and outlines the articles V.1–V.5. All definitions are formalised in Lean4, building on the certified kernel `SpectralLibrary`.

1.1 The magnet metaphor

Imagine a vast space containing an enormous number of points – each point represents a candidate pair (p, q) . The lantern (ground state) is constructed so that the only bright points correspond to actual Goldbach pairs. The four pillars ensure that the light spreads quickly, that the image is compressible, and that we can read the brightest point – a prime pair – in polynomial time.

2 Recap of the spectral reduction lemma

The Spectral Reduction Lemma (Article0.8) states that any problem that can be faithfully translated into a spectral Hamiltonian $H = \hat{V} + L$ on the hypercube (or a bounded-degree graph) satisfying the four pillars is solvable in polynomial time. The four pillars are:

1. **P1 (Perron-Frobenius)**: Unique strictly positive ground state ψ_0 .
2. **P2 (Kithima Bridge)**: Constant spectral gap $\gamma(H) \geq 2$ (polynomial is sufficient).
3. **P3 (Logarithmic area law)**: Entanglement entropy $S(\rho_B) = O(\log M)$, hence MPS representation with bond dimension polynomial in M .
4. **P4 (Deterministic extraction)**: D-RSP algorithm extracts a satisfying assignment (or proves unsatisfiability) in polynomial time.

For Goldbach, we use the CD strategy: we directly encode the problem as a SAT instance and rely on the generic SAT solver.

3 Structure of the series V

The series V consists of the following articles:

Article	Title	Main content
V.0	Foundations of the Spectral Proof of Goldbach’s Conjecture	This article: introduction, plan, recall lemma.
V.1	Encoding Goldbach as a Spectral Hamiltonian	Configuration space (hypercube), diagonal potential (zero on Goldbach pairs, n^2 elsewhere), hypercube Laplacian, irreducibility.
V.2	Pillar P1 – Uniqueness and Positivity	Perron-Frobenius theorem applied to transition matrix, unique strictly positive ground state.
V.3	Pillar P2 – The Kithima Bridge for Goldbach	Conductance bound, Kithima Bridge inequality, polynomial spectral gap.
V.4	Pillar P3 – Logarithmic Area Law and MPS	Exponential decay of correlations, Brandão–Horodecki, Hilbert curve, renormalisation, MPS bond dimension.
V.5	Pillar P4 – Deterministic Extraction via D-RSP	Power iteration on MPS, amplitude greedy bit extraction, polynomial-time algorithm.
V.6	Synthesis – Goldbach’s Conjecture Resolved	Correspondence dictionaries, integration, the 34-problem corpus.

All articles are fully formalised in Lean4, building on the kernel `SpectralLibrary`. No unproven assumptions remain.

4 Formalisation in Lean4 (overview)

The master file for Series V will be `MainV.lean`. It imports the results of V.1–V.5 and states the final theorem.

Listing 1: `MainV.lean` (sketch)

```

1 import V.V1
2 import V.V2
3 import V.V3
4 import V.V4
5 import V.V5
6
7 open SpectralLibrary
8
9 -- Final theorem: Goldbach’s conjecture
10 theorem goldbach_conjecture (n : ℕ) (heven : Even n) (hgt : n > 4) :
11   ∃ p q : ℕ, Prime p ∧ Prime q ∧ p + q = n :=
12   goldbach_spectral_proof
13
14 -- Axiom audit: only standard axioms
15 #print axioms goldbach_conjecture

```

5 Conclusion

This article sets the stage for a complete spectral proof of Goldbach’s conjecture using the constructive direct strategy. The subsequent articles (V.1–V.6) will provide the detailed construction of the circuit, the Hamiltonian, the verification of the four pillars, and the extraction algorithm. The proof is unconditional and fully formalisable in Lean4. Once completed, Goldbach’s conjecture will become another jewel in the crown of the spectral reduction lemma.

References

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